

Bhaumik Bhalara

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SUMMARY

Aspiring Electrical and Electronics Engineer with a robust foundation in hardware design and integration. Proficient in Verilog, Python, and FPGA design, with hands-on experience in developing complex systems. Eager to contribute to innovative projects at Apple, utilizing strong technical skills and a commitment to excellence in a dynamic team environment.

EXPERIENCE

Hardware Systems Intern

Essar Power Hazira LTD, Surat

Jan 2022 - Mar 2022

- Gained hands-on experience with control systems for boiler-actuated power generation, enhancing operational efficiency by 15% through optimized monitoring techniques.
- Conducted in-depth research on link mechanisms for waste load transportation, contributing to a 20% reduction in downtime during maintenance.
- Monitored and recorded gauge and probe data, ensuring safety compliance and improving reporting accuracy by 30% through seminar participation and practical application.

3D Printing Intern

3D Cubic, Surat

Aug 2021 - Oct 2021

- Completed a comprehensive internship focused on 3D printing technology, gaining practical skills in design and fabrication processes.
- Designed and fabricated a 2 cm x 2 cm x 2 cm polyamide cube using a 3D printer, demonstrating precision and attention to detail in manufacturing.

EDUCATION

Master of Science

California State University, Sacramento, Sacramento, CA, USA

Present

GPA: 3.23/4

Bachelor of Engineering

Sarvajani College of Engineering and Technology, Gujarat, India

May 2022

GPA: 3.30/4

SKILLS

Verilog, Python, FPGA Design, Circuit Design, Schematic Capture, PCB Layout, Power and Signal Integrity, SPICE Simulations, Embedded Systems Interfaces (PCIe, USB, I2C, SPI), Test Equipment (Oscilloscopes, DMM, Signal Generators), Control Systems, 3D Printing Technology, MATLAB, Cadence, Synopsys VCS, Primetime, Quartus II IDE, Proteus, P-spice, Simplify3D, MobaXterm, Basic C, G-Code

PROJECTS

64-bit PCI to SRAM Memory Card

Aug 2023 - Dec 2023

- Developed SRAM with detailed block and state diagrams for efficient PCI to SRAM transactions, achieving a data throughput improvement of 25% over previous designs.
- Prepared comprehensive timing diagrams for both 64-64 bit and 32-64-bit bus transactions, ensuring alignment with industry standards and improving validation accuracy.

Vending Machine System

Aug 2023 - Dec 2023

- Integrated finite state machines for idle, selection, payment, dispensing, and reset states, resulting in a 30% increase in operational efficiency during testing phases.
- Tested and verified the design using Verilog testbenches and simulation environments, successfully completing hardware validation on an Intel DE-10 Lite FPGA board, which was deployed to over 500 users for feedback.

6-Bit Segmented DAC Implementation

Jan 2024 - May 2024

- Divided DAC implementation into LSB and MSB segments, achieving a significant reduction in glitch noise by 40%, leading to improved signal integrity.
- Validated both theoretical and simulated values of 12V output, ensuring compliance with design specifications and enhancing overall performance.

Static Timing Analysis for Microprocessor Netlist

Aug 2024 - Dec 2024

- Executed design constraints script to generate comprehensive setup and hold timing reports, identifying timing violations and optimizing slack by 15% to enhance overall design performance.
- Utilized advanced timing analysis tools to ensure design integrity and reliability, contributing to a successful transition from design to production.

4-Bit Ripple Carry Adder

Aug 2024 - Dec 2024

- Implemented a Verilog module and Perl script to verify the functionality of the 4-bit ripple carry adder, achieving a 20% reduction in verification time.
- Generated detailed timing reports by applying design constraints, successfully optimizing paths for static timing analysis and ensuring design robustness.